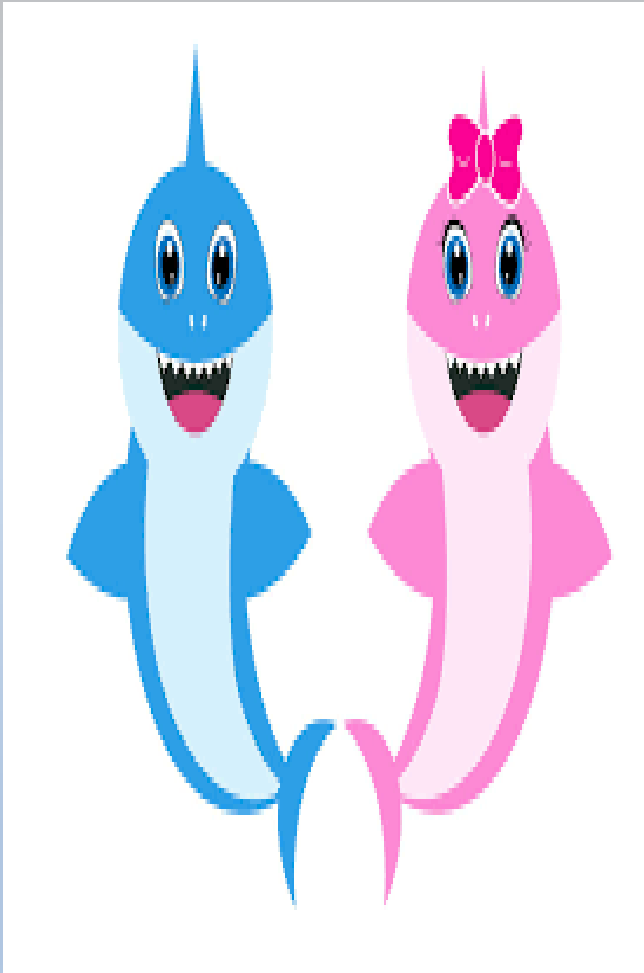


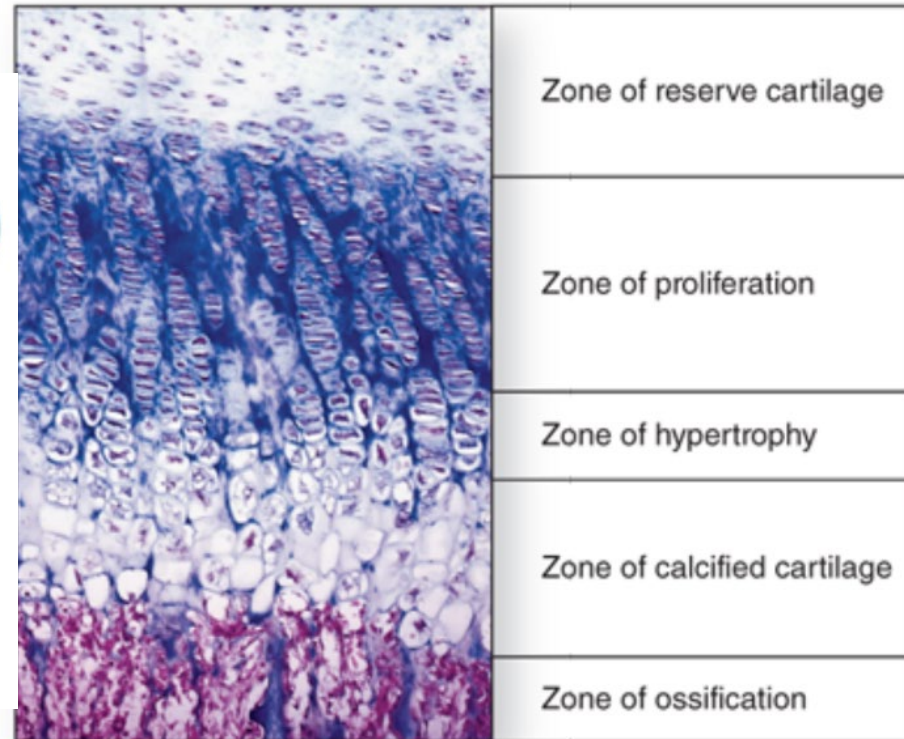
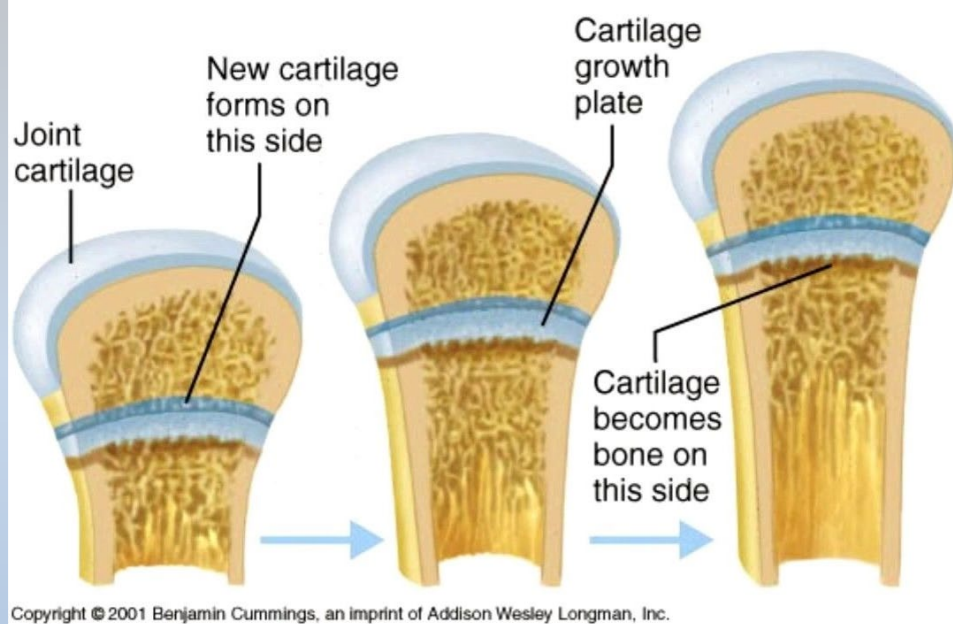
Session Objectives

1. Discuss the differences in pediatric bone structure and how this affects disease processes
2. Discuss bone fractures in children and how to use the Salter Harris guide for fractures involving the growth plate
3. Discuss the most common developmental and acquired deformities of foot and hip how to diagnose based on age and presentation
4. Review presentation, cause, diagnosis and treatment of pediatric bone and joint infections
5. Review other osteopathic disorders- rickets and Transient Synovitis
6. Identify signs of non-accidental trauma

Baby Shark?Not so far off from truth



Epiphyseal Plate



b Epiphyseal plate

Source: Anthony L. Mescher: Junqueira's Basic Histology, 14th Edition.
www.accessmedicine.com
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Pediatric Bone Injuries

- Common
- Related to brain and motor development
- Location based on age
- Intentional vs. Unintentional



Salter-Harris Classification of Physeal Fractures



Injuries Involving the Epiphyseal Plate

BY ROBERT B. SALTER, M.D., F.R.C.S.(C)*, AND W. ROBERT HARRIS, M.D., F.R.C.S.(C)†,
TORONTO, ONTARIO, CANADA

An Instructional Course Lecture, The American Academy of Orthopaedic Surgeons

Injuries involving the epiphyseal plate present special problems in diagnosis and management. The dread complication of serious disturbance of growth is usually predictable and, in certain circumstances, can be prevented. Thus, knowledge of the prognosis for a given injury to the epiphyseal plate in a particular child is of considerable importance to the surgeon, who has the dual responsibility of treating the child and advising the parents. The purpose of this presentation is to discuss epiphyseal-plate injuries from both the clinical and the experimental points of view.

J Bone Joint Surg Am. 1963;45:587-622.
ROBERT B. SALTER and W. ROBERT
HARRIS

Salter Harris Type 1

Growth plate

- Fracture through the zone of hypertrophy
- Width of the physis increases.
- **Growth disturbances are uncommon**
- **Treat with casting or splinting**



Salter Harris Type 2

Metaphysis and growth plate

- **Most common type - 75%**
- Usually will not cause functional limitations.
- Occur above the growth plate



Salter Harris Type 3

Epiphysis and growth plate

- Potentially damaging
- Chronic disability due to **extension of fracture into the articular surface**
- Arthritis possible without reduction – surgery
- Treatment surgical.



Salter Harris Type 4

Epiphysis, growth plate and metaphysis is

- Intra-articular fracture
- May cause chronic pain
- May cause deformity of the joint
- Surgery needed for repair



Salter Harris Type Five

Compression or Crush Fracture

- Rare
- Growth affected
- Injury occurs at the growth plate
- X-ray findings are subtle/difficult to see
- High degree of suspicion needed
- Most dx made in retrospect after growth plate sealed



Diagnosing DDH

Diagnosis made with high degree of suspicion

- By special physical exam tests
- By ultrasound

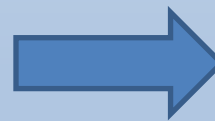
– Tests to Detect DDH:

- **Barlow Test:** provoke hip prone to dislocation
- **Ortolani Test:** move a dislocated hip back into acetabulum
- **Galeazzi Test:** evaluates for current hip dislocation



Barlow

musculoskeletalkey.com



Ortolani



Surgery vs. Bracing

Treatment Goals: relieve symptoms, restore ROM, and containment of the hip.

- Minimal Weight bearing
- Pain control
- Physical therapy
- Bracing (Petrie Cast) if needed to help contain femoral head within acetabulum
- If severe, surgery will be required



<https://www.pinterest.com/pin/64387469644844752/>

Slipped Capital Femoral Epiphysis

Obese, pubertal children during their growth spurt

- Mean age girls: 12
- Mean age boys: 13.5
- ~2x more common in **boys**
- May have positive family history



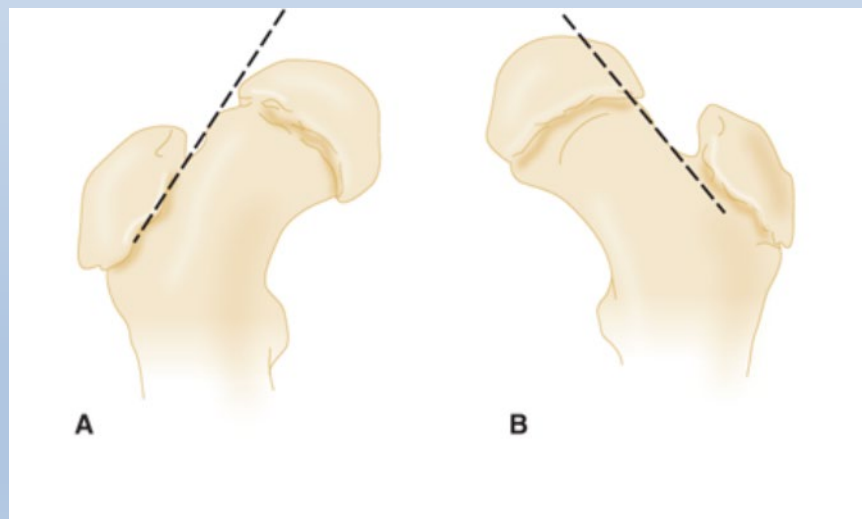
Signs & Symptoms:

- Hip, thigh, groin, or knee pain
- Limp
- Externally rotated leg at rest
- **External rotation with flexion**
- Decreased ROM (Flexion and IR)



Slipped Capital Femoral Epiphysis

- **Diagnosis** via X-ray AP and Frog Leg View
 - AP: **Klein's line** and **Trethowan's sign** (A)



Source: J.E. Tintinalli, J.S. Stapczynski, O.J. Ma, D.M. Yealy, G.D. Meckler, D.M. Cline:
Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th Edition
www.accessmedicine.com
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Metatarsus Adductus

Line through heel should cross through second toe

In metatarsus adductus, line will cross lateral to second toe.

Treatment involves stretching forefoot

Parents can do at home

Consider referral if not better by eight months





Infection of Bones and Joints

Pathogenesis- how are infections acquired
and properties unique to bone/joint

Presentation- PE and Lab findings

Bacteriology- most common causes

Management/Treatment



Acute Osteomyelitis

Who gets it?

1 in 5000 under 13 years

Most cases are in young children

Increased risk in sick children and SCD

Where does it occur?

Metaphyseal end of long bones

Femur, Tibia and Humerus - 79%

Risk related to:

Bones more porous

Higher rates of blood infections

Higher rates of accidents



Treatment

Treatment for osteomyelitis always starts with IV therapy for a minimum of 2-3 weeks

Treatment starts as “empiric”- best educated guess at bacteria then follows with culture results

Increasing MRSA has guided empiric tx to clindamycin-vancomycin- linezolid first line



Pathogenesis

Ditto to osteomyelitis

Three different types of inoculation

Bacteriology similar

Elevated WBC and inflammatory markers

Knee > Hip > Ankle = 80% of infections



Transient (toxic) Synovitis of the Hip

Common- Acute

Must be differentiated from Septic Hip

Exact etiology unclear

Generally seen ages 3-10

Preceding viral/GI infection common

Exam of hip shows pain with passive hip motion and +/- limp-

Fever – if present low grade



TS Diagnosis and Treatment

Diagnosis is clinical

Rest

Heat

Massage

NSAIDs

Re-evaluate in 24-48 hours

Return to full activity 7-10 days



Transient vs. Septic Hip

Transient Synovitis

+/- Fever

Subtle onset

+/- Limp

Normal ESR

Prior illness

Normal x-ray/US

Bacterial Septic Arthritis

High fever

Sudden onset

Limp present

ESR over 40

- Prior illness

Abnormal x-ray/US

Rickets

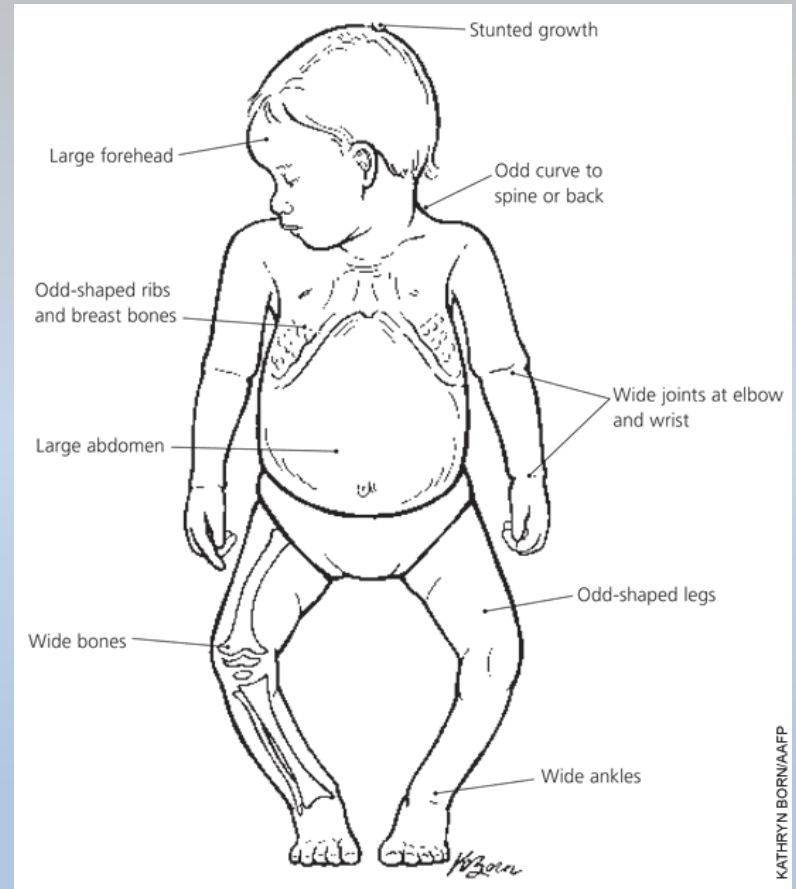
Disorder of Ca & PO₄ homeostasis

Severe prolonged Vit D deficiency

Bones softer & weaker

Nutritional deficiency most common

Chronic diseases may also cause



Summary

1. The bones of children differ from adults and results in different types of fractures and fractures that affect the growth plate
2. Knowing the signs of accidental vs. non-accidental trauma in children is important
3. Disorders of the hip can be differentiated based on age and presentation symptoms
4. Bone and joint infections share many similarities and are both considered emergent conditions to diagnose and treat
5. Foot malformations are common in children